

Donguk **Shin**

sdg1225@pusan.ac.kr

Education

Pusan National University Busan, Republic of Korea — *March 2021 – Present*
Bachelor of Business Administration — *(Expected Graduation: February 2028)*

- Double Major in Fintech Convergence Major
- Total GPA: 4.30 / 4.50 (Rank: 12 / 253)
  Major GPA: 4.41 / 4.50

University of Jyväskylä Jyväskylä, Finland — *August 2025 – December 2025*
Exchange Program, School of Business and Economics

Projects

AI-Powered Parametric Insurance for Smart Farm — *April 2026 – May 2026*

National University Student Risk Management Competition | Team member

- **Objective & Outcome:** Aimed to establish a proactive safety net for smart farm complex risks, resulting in the development of a dual-engine parametric insurance.
- **Role:** Architected an end-to-end data pipeline processing 12.47 million records to quantify environmental risks and calculate parametric insurance premiums.
- Engineered a 7-day predictive warning model using LightGBM via sliding-window feature engineering and out-of-sample validation, achieving a 74% F1 score.

Knowledge Graph-Based B2B Matching System Implementation — *June 2025 – July 2025*

Industry-Academic Collaboration Project with Illunex | Team member

- **Objective & Outcome:** Aimed to infer potential B2B demand-supply associations in the absence of explicit transaction records, resulting in the successful development of an automated matching system leveraging Qdrant vector databases and RAG technologies.
- **Role:** Integrated corporate metadata and derived categorical variables through text normalization to build a robust data foundation.
- Architected a Knowledge Graph of 9 node types and 49,000+ relationships, optimizing GNN embeddings with TF-IDF similarity edges.

RMR Recipe Database Construction — *May 2025 – July 2025*

Youth Work Experience Project supervised by WAVE & VIBE | Team member

- **Objective & Outcome:** Aimed to maximize RMR production profitability by developing alternative recipe algorithms, resulting in the successful deployment of a standardized recipe database on AWS RDS MySQL.

- **Role:** Separated ingredients from seasonings and standardized measurement units into a uniform gram metric from 170,000 unstructured recipes.
- Mapped irregular ingredient terminology to official national food standards using NLP sentence embeddings and cosine similarity thresholds.

Optimal Location Selection for Elderly Welfare Facilities *March 2025 – June 2025*

Term Project for course Data Science Introduction of 2025 – 1st semester | Personal Project

- **Objective & Outcome:** Aimed to resolve regional welfare infrastructure imbalances in a super-aged society by identifying optimal locations for new elderly welfare facilities.
- Mapped the city into Uber's H3 Hexagon Grid System for granular analysis and resolved missing coordinate data by developing a custom 3-step geocoding pipeline via KakaoMap API.
- Designed a location suitability model evaluating 4 core metrics—demand, supply, accessibility, and economy—via Min-Max normalization, successfully pinpointing the top 3 optimal sites.

Internships

Smart Factory Laboratory, Pusan National University *March 2026 – Present*

Undergraduate Research Assistant (Advisor: Professor Jun-hee Han, Dept. of Industrial Engineering)

- Acquiring academic methodologies and foundations in preparation for graduate studies in Operations Management and Industrial Engineering.
- Currently, researching scheduling that considers the mental workload of workers when working with robots.

Work-Study Student *2021 2nd, 2024 1st and 2025 1st Semester*

Assistant in the administrative and office of the business school

- Provided administrative support and assisted in key department events, including university admissions and major examinations.

Experiences

IBA, Intelligent Business Analytics Club *March 2026 – Present*

Club Executive / Treasurer

- Manage the analytics organization's overall budget, plan educational curriculums, and coordinate operations with internal and external institutions.
- Elected to serve as the President for the upcoming semester.

Busan Youth Volunteer Group *March 2026 – Present*

Volunteer tutor & Operations Staff

- Designed and led digital literacy programs for the elderly as part of a municipal youth volunteer initiative, focusing on self-service kiosk navigation and voice phishing prevention.
- Participated in the end-to-end project lifecycle, managing partner institution outreach, budget allocation, material development, and the execution of educational sessions.

PNU Biz Global Tutoring Program *May 2025 – June 2025*

Peer Tutor

- Selected as an academic mentor for an international student within the School of Business Administration.
- Conducted tailored tutoring sessions and designed specialized mock examinations for a Kazakhstani student, significantly improving her comprehension and performance in the 'Operations Management' course.

TDF_Busan, Volunteer Association *March 2025 – June 2025*

Volunteer Tour Guide & Operations Staff

- Conducted English-guided cultural tours for international tourists visiting Busan, enhancing global communication competencies.
- Managed, recorded, and official volunteer performance logs and administrative documents within the volunteer association.

IBA, Intelligent Business Analytics Club *August 2024 – July 2025*

Active member

- Completed an intensive 6-month curriculum covering fundamental statistics and Python-based data pipelines (collection, preprocessing, EDA, and modeling).
- Executed diverse data-driven projects using machine learning to solve real-world business challenges, including the development of a used car price forecasting model and a financial credit rating classification system.

Certifications & Qualifications

Certifications

ADsP (Advanced Data Analytics Semi-Professional) *March 2025*

SQLD (SQL Developer) April *2025*

Qualifications

TOEFL iBT: 91 *September 2024*

Honors & Awards

Honors

Busan Regional Talent Scholarship *2025 – 2nd Semester*

Busan

Fintech Convergence Major Scholarship *2024 – 2nd & 2025 – 1st Semester*

Pusan National University

Business Administration Endowed Scholarship *2024 – 2nd Semester*

Pusan National University

Awards

National University Student Risk Management Competition, Encouragement Award *May 2026*

Samsung Fire & Marine Insurance

Fintech Research Idea Competition, Excellence Award *February 2025*

Pusan National University